



Preventing Cryptocurrency Scams through Automated and Manual Moderation

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Problem Description

Cryptocurrency scams are among the fastest-growing forms of fraud, with reported losses increasing roughly 53% to nearly \$3.9 billion in the past year, according to the FBI. This increase can be attributed to the lack of regulation in this space, the ability to remain anonymous and the permanent nature of transactions. This allows funds to be transferred without oversight, making it incredibly challenging for victims to seek retribution. Bad actors predominantly target new investors and young individuals who are interested in cryptocurrency or elderly individuals who are less tech-savvy. To combat this, our platform intends to minimize fraud by providing robust moderation of such scams and a comprehensive reporting system through DMs.

Our Cryptocurrency Scam Policy

We are committed to fostering a safe and trustworthy community for all our users. To uphold this commitment, we have implemented a strict policy to combat cryptocurrency-related scams, which pose a threat to our users' assets and personal safety.

- Considering scamming schemes that are often used in cryptocurrency scams, we flag the following cryptocurrency-related abuse types:
- **Suspicious Links:** Any links that may lead to a disreputable site
 - **Blackmail:** Attempts to threaten users into providing cryptocurrency
 - **Investment Scams:** Deceptive schemes promising high returns on investments
 - **Imminent Danger:** Any situation where users' personal safety is at immediate risk

- We employ a multi-step approach to identify and mitigate cryptocurrency scams:
- **Automated Detection:** Automatically flagging content that may violate our policies
 - **User Reporting:** Our users can report and classify any potentially violating content
 - **Human Moderation:** Our team of moderators conducts manual reviews to ensure legitimate reports and swift action is taken against fraudulent activities

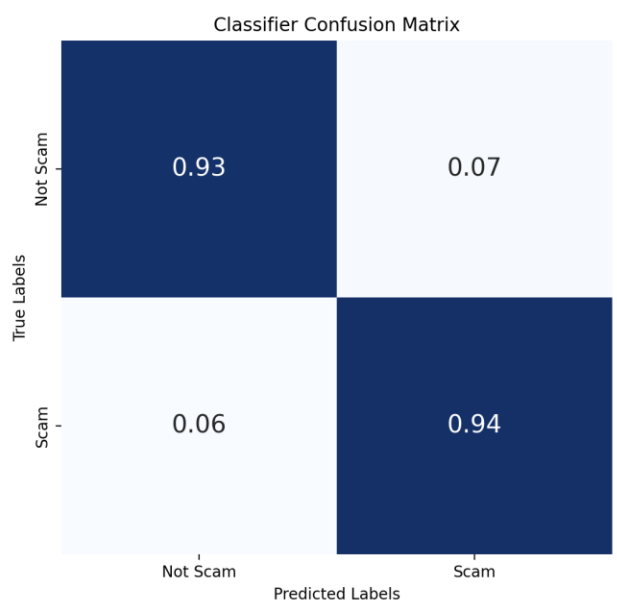
- In the event of a policy violation, we may take the following actions:
- **Content Removal:** Immediate removal of the violating message
 - **Account Probation:** Additional surveillance of violator to make sure repeated violations are flagged – violator's access to platform features is limited
 - **Report Account to Discord:** Escalate report to Discord to investigate violator's history in other channels as well and decide whether to take further action
 - **Banning Account:** Permanent banning of violator's account from the server
 - **Contact Local Authorities and Our Legal Team:** Involve law enforcement in cases of severe violations of our policy - reported content may be provided during investigation

Evaluation

Cryptocurrency Scam Classifier

Quantitative Analysis:

- Confusion matrix generated from test set evaluation indicates low false positive and false negative rates



Qualitative Analysis

- Sent messages that employed various cryptocurrency scam tactics as well as messages containing innocuous, cryptocurrency-related discourse
- Results:
 - 5/6 of scam messages were classified correctly
 - 5/7 of non-scam messages were classified correctly
 - Results align with quantitative results: false positive rate is tolerable but to ensure scams are not left unnoticed, need to reduce false negative rate
- Examples of false positives:
 - "Join our webinar on crypto investment strategies this Friday at 3 PM EST."
 - "Introducing our new feature: instant crypto-to-fiat conversions on our platform."
- Example of false negative:
 - "We're giving away 5 ETH to the first 100 people who send 0.05 ETH to this wallet: 0x4eE7..."

- Main takeaways:
 - Our model performs moderately well in classifying both scam and non-scam messages
 - However, it is overestimating the link between the urgency/immediacy of a message and how scam-like the message is
 - Both the false positives relate to claims that were near-term or instantaneous

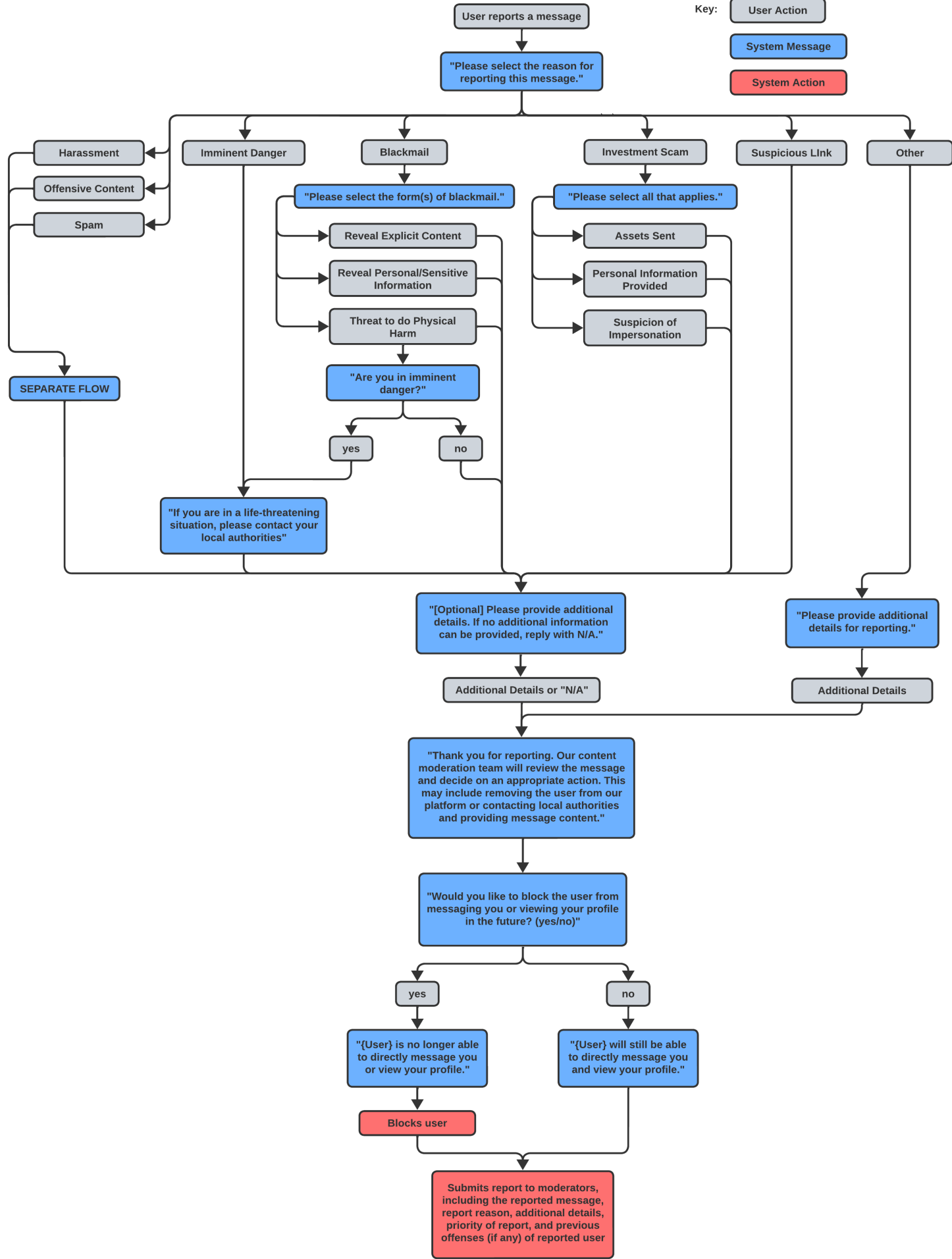
VirusTotal

- Pros: provides broad detection as it leverages multiple scanners/services, APIs are easy to use and are free
- Cons: results are sometimes inconsistent, requires providing exact url (variations are undetectable)

Technical Back-End

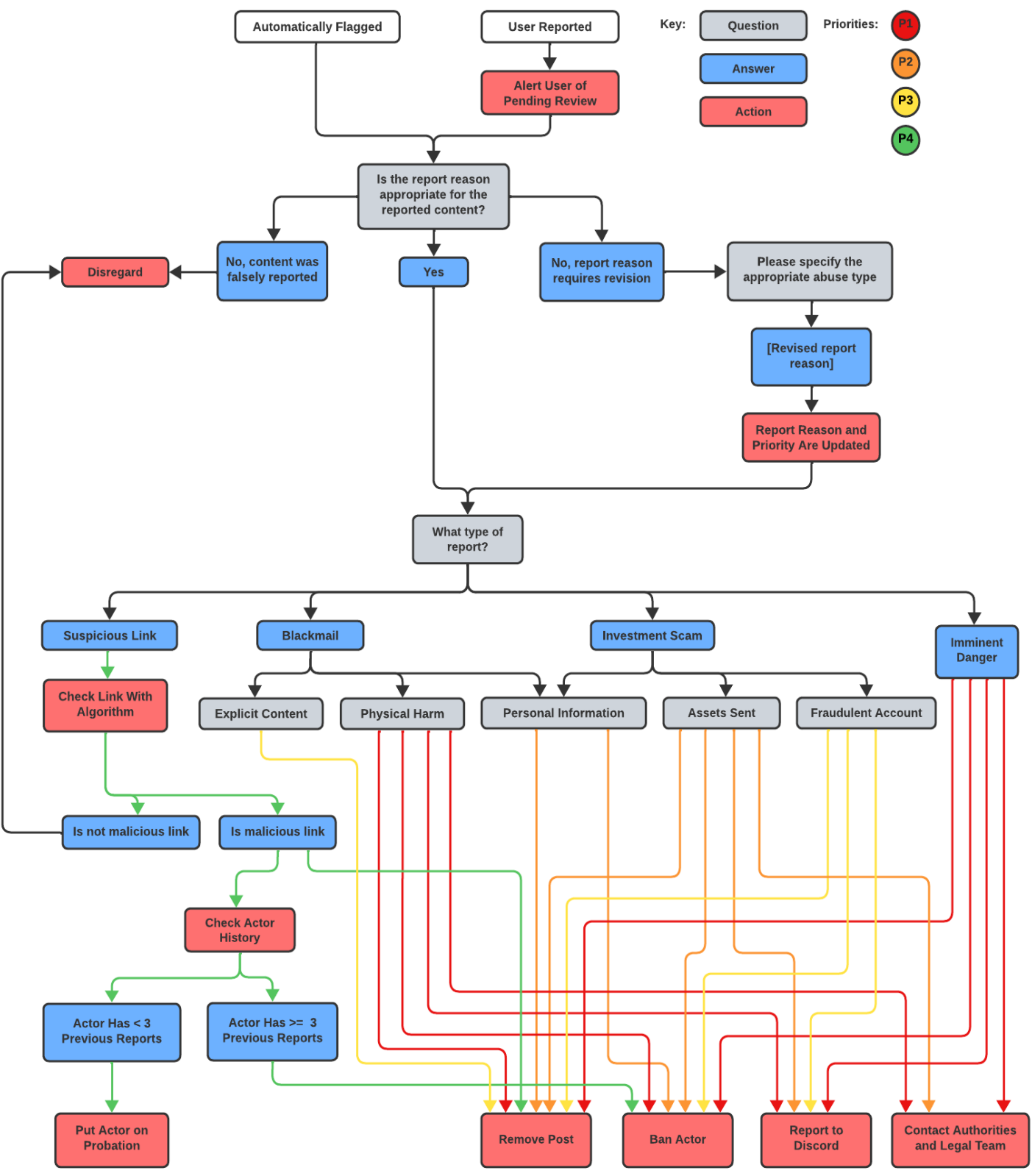
User Reporting Flow

- "Other" and the ability to specify additional details are presented as options to cover any other forms of harassment or fraudulent activities not captured by the categories we provide



Moderator Flow

- Actions are preselected to increase moderator velocity – moderators can confirm or revise actions
- User database: statistics such as the number of previous reports made against the reported user are provided with the report to aid in decision making
- Reports are reviewed based on priority - abuse types that involve physical harm/immediate danger to the user are prioritized first to ensure user/community safety



Looking Forward

Our implementation of both automated and manual moderation of cryptocurrency-related scams and a comprehensive reporting system significantly enhances platform safety by reducing occurrences of such scam incidents, increasing user trust, and ensuring fast response times to reports. To further improve our design goal, we propose the following changes going forward:

- Launch campaigns and provide tutorials to educate users about scam tactics and safety practices
- Implement real-time alerts for users about emerging scam trends and threats
- Develop an AI-powered chatbot to assist users in reporting scams that don't fit a framework and to provide immediate advice
- Allow users to customize their security settings and scam detection sensitivity
- Scale up our training dataset, supplementing with messages from online recounts of scams to improve classifier performance
- Implement multilingual support
- Enhance suspicious link detection to be resilient against adversarial formatting, such as special symbols